Thermalizing Quantum Systems & Evaluating Partition Functions with a Quantum Computer

Pawel M. Wocjan

School of Electrical Engineering and Computer Science

University of Central Florida

Orlando

wocjan@eecs.ucf.edu

Joint work with

- David Poulin (University of Sherbrooke)
  arXiv preprint 0905.2199

- Chen-Fu Chiang (University of Central Florida)

- this work has been supported by the National Science Foundation grants CCF-0726771 and CCF-0746600

Definitions

- let H be a Hamiltonian with spectral decomposition

$$H = \sum_{a=1}^{D} E_a |\psi_a\rangle\langle\psi_a| \,, \qquad E_a \in [0,1]$$

- the partition function Z_β at inverse temperature β is

$$Z_\beta := \sum_{a=1}^{D} e^{-\beta E_a}$$

- the thermal state ρ_β at temperature β is

$$\rho_\beta := \sum_{a=1}^{D} \frac{e^{-\beta E_a}}{Z_\beta} |\psi_a\rangle\langle\psi_a|$$

Tasks

- Evaluating parition functions

  given H, β, and ϵ, output a random value $\tilde{Z}_\beta$ such that

$$\Pr\left(\left|\tilde{Z}_\beta - Z_\beta\right| < \epsilon Z_\beta\right) > \frac{3}{4}$$

- Thermalizing quantum systems

  given H, β, and ϵ, prepare a quantum state $\tilde{\rho}_\beta$ with

$$\frac{1}{2}\|\rho_\beta - \tilde{\rho}_\beta\|_{\mathrm{tr}} \leq \epsilon$$

Thermalizing Quantum Systems

- given H, β, and ϵ, prepare a quantum state $\tilde{\rho}_\beta$ with

$$\frac{1}{2}\|\rho_\beta - \tilde{\rho}_\beta\|_{\mathrm{tr}} \leq \epsilon \quad \text{(*)}$$

- the above task was studied by Terhal and Divincenzo in Phys. Rev. A 61, 022301
  - constructed a TCP whose fix-point is the thermal state ρ_β in an idealized setting (perturbation theory)
  - did not prove how the number of iterations has to be chosen to obtain a state $\tilde{\rho}_\beta$ satisfying (*)
  - numerical simulations suggests that this algorithm may outperform classical algorithms

Our Quantum Algorithm

- given H, β, and ϵ, our quantum algorithm makes it possible to prepare $\tilde{\rho}_\beta$ with

$$\frac{1}{2}\|\rho_\beta - \tilde{\rho}_\beta\|_{\mathrm{tr}} \le \epsilon$$

- its running time is

$$\sqrt{\frac{D}{Z_\beta}} \cdot poly(\log D, 1/\epsilon, \beta)$$

Classical Hamiltonians

- let Ω be a set whose elements x correspond the states of some classical system
- let $E : \Omega \to [0, 1]$ denote the energy function, assigning to each state x its energy $E(x)$
- let

$$H = \sum_{x \in \Omega} E_x |x\rangle\langle x|$$

- thermalizing such classical systems is easier because H is diagonal in the computational basis $\{|x\rangle\}_{x\in\Omega}$

Metropolis Algorithm

- given such classical Hamiltonian H, we can always construct a Markov chain P whose limiting distribution is equal to ρ_β
- the spectral gap δ determines how fast we can approach the Boltzmann distribution
- the run time scales like $1/\delta$
- often, it is not possible to determine (or bound) the spectral gap $\Rightarrow$ heuristics

Quantized Metropolis Algorithm

- given such classical Hamiltonian H, there is a "quantized" Metropolis algorithm preparing $|\tilde{\varphi}_\beta\rangle$ with

$$\big\| |\varphi_\beta\rangle - |\tilde{\varphi}_\beta\rangle \big\| \leq \epsilon$$

  where $|\varphi_\beta\rangle$ is a "quantized" version of the Boltzmann distribution

$$|\varphi_\beta\rangle := \sum_{x\in\Omega} \sqrt{\frac{e^{-\beta E_x}}{Z_\beta}} |x\rangle$$

- the run times scales like $\sqrt{1/\delta}$
- proved in Somma, Boixo, Barnum, and Knill, PRL 101, 130504 and W. and Abeyesinghe, PRA 78, 032311

Grover's State Engineering

- prepare the state

$$\frac{1}{\sqrt{D}} \sum_{x \in \Omega} |x\rangle \otimes |E_x\rangle \otimes |0\rangle$$

- apply a controlled rotation to obtain the state

$$\frac{1}{\sqrt{D}} \sum_{x \in \Omega} |x\rangle \otimes |E_x\rangle \otimes \left(\sqrt{e^{-\beta E_x}} |0\rangle + \sqrt{1 - e^{-\beta E_x}} |1\rangle \right)$$

Grover's State Engineering

- apply Grover to increase the overlap with the $|0\rangle$ component and measure in the basis $|0\rangle$, $|1\rangle$
- if we obtain $|0\rangle$, then the post-measurement state is

$$\sum_{x\in\Omega}\sqrt{\frac{e^{-\beta E_x}}{Z_\beta}}|x\rangle\otimes|E_x\rangle\otimes|0\rangle$$

- the expected run time to obtain this state scales like

$$\sqrt{\frac{D}{Z_\beta}}$$

Our Quantum Algorithm

- is a generalization of the algorithm based on Grover's state engineering to quantum Hamiltonians
- let H be an arbitrary Hamiltonian with spectral decomposition

$$H = \sum_{a=1}^{D} E_a |\psi_a\rangle\langle\psi_a| \,, \quad E_a \in [0,1]$$

Error Sources

- given H, β, and ϵ, our quantum algorithm makes prepares $\tilde{\rho}_\beta$

$$\frac{1}{2}\|\rho_\beta - \tilde{\rho}_\beta\|_{\mathrm{tr}} \leq \epsilon$$

- the errors come from three sources:

$$\|\rho-\tilde{\rho}\|_{\mathrm{tr}} \leq \|\rho-\rho_{\text{sim}}\|_{\mathrm{tr}}+\|\rho_{\text{sim}}-\rho_{\text{prec}}\|_{\mathrm{tr}}+\|\rho_{\text{prec}}-\rho_{\text{fail}}\|_{\mathrm{tr}}+\|\rho_{\text{fail}}-\tilde{\rho}\|_{\mathrm{tr}}$$

  - ρ_{sim} imperfect simulation of $\exp(iH)$
  - ρ_{prec} limited precision of phase estimation
  - ρ_{fail} nonzero failure probability of phase estimation

Imperfect Hamiltonian Simulation

- assume that H is sparse and let $U = \exp(iH)$
- using techniques for simulating Hamiltonian time evolutions, we can implement U_{sim} with

$$\|U - U_{\text{sim}}\| \leq \epsilon_{\text{sim}}$$

- the run time scales like

$$poly(logD, 1/\epsilon_{\text{sim}}, s)$$

Technical Arguments

- matrix logarithm is Lipschitz continuous:

  $$\|U - U_{\text{sim}}\| \leq \epsilon_{\text{sim}}$$

  implies that there is a Hamiltonian H_{sim} with

  $$\|H - H_{\text{sim}}\| \leq O(\epsilon_{\text{sim}})$$

- thermal states are Hölder continuous:

  $$\|H - H_{\text{sim}}\| \leq O(\epsilon_{\text{sim}})$$

  implies that the corresponding thermal states satisfy

  $$\|\rho - \rho_{\text{sim}}\|_{\text{tr}} \leq O(\sqrt{\beta \epsilon_{\text{sim}}})$$

- we assume $U_{\text{sim}} = U$ (and keep track of the error)

Maximally Entangled State

- let

$$|\Phi\rangle := \frac{1}{\sqrt{D}} \sum_{a=1}^{D} |i\rangle \otimes |i\rangle$$

- let

$$V = \sum_{a=1}^{D} |\psi_a\rangle\langle a|$$

- it is important that we can express $|\Phi\rangle$ as

$$|\Phi\rangle = \sum_{a=1}^{D} |\psi_a\rangle \otimes |\psi_a^T\rangle$$

this follows from $(V \otimes I_D)|\Phi\rangle = (I_D \otimes V^T)|\Phi\rangle$

Phase Estimation with Boosting

- recall that $U = \exp(iH)$
- $\Rightarrow$ we have

$$(U \otimes I_D)|\psi_a\rangle \otimes |\psi_a^T\rangle = e^{iE_a}|\psi_a\rangle \otimes |\psi_a^T\rangle$$

$$(U \otimes I_D)|\Phi\rangle = \sum_{a=1}^{D} e^{iE_a}|\psi_a\rangle \otimes |\psi_a^T\rangle$$

- $\Rightarrow$ the energy levels E_a of H are mapped bijectively onto the phases e^{iE_a} of U
- $\Rightarrow$ apply phase estimation to estimate the energy levels

Phase Estimation with Boosting

- use phase estimation with $O(1/\epsilon_{\text{prec}})$ ancillas and run $O(\log(1/\epsilon_{\text{fail}}))$ boosting rounds
- $\Rightarrow$ this transforms $|\Phi\rangle$ into the state

$$\sum_{a=1}^{D} |\psi_a\rangle \otimes |\psi_a^T\rangle \otimes \left(c_a^-|E_a^-\rangle + c_a^+|E_a^+\rangle\right) + |\text{junk}\rangle$$

where

- $E_a^+ - E_a \leq \epsilon_{\text{prec}}, \qquad E_a - E_a^- \leq \epsilon_{\text{prec}}$
- $|c_a^+|^2 + |c_a^-|^2 = 1$
- $\| |\text{junk}\rangle \| \leq \epsilon_{\text{fail}}$